

Z 9/1 gramofonska ploča

$$V = 33/\text{min}$$

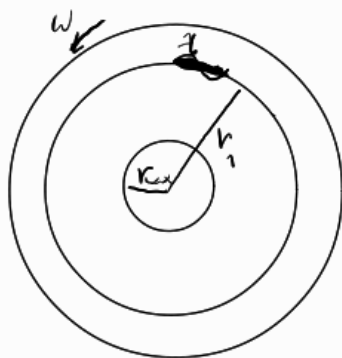
$$r_1 = 10 \text{ cm} \rightarrow V_1 = 440 \text{ s}^{-1}$$

$$\lambda_1 = ?$$

$$r_{\text{max}} = 3 \text{ cm}$$

$$d = 20 \mu\text{m}$$

$$V_{\text{max}} = ?$$



$$V_1 = 2\pi r_1 V = \lambda_1 \cdot V_1 \quad \leftarrow \begin{matrix} t_{\text{osc}} \\ \lambda_1 \end{matrix}$$

$$\lambda_1 = 2\pi r_1 \frac{V}{V_1} = 2\pi \cdot 0,1 \text{ m} \cdot \frac{9,55/\text{s}}{440/\text{s}} = \underline{\underline{0,8 \text{ mm}}}$$



$$\lambda_{\text{max}} = d$$

$$V_{\text{max}} = \frac{2\pi r_{\text{max}} V}{d} = 5,2 \text{ kHz}$$

9.2

$$y(x, t) = a \sin(kx - \omega t - \frac{\pi}{4})$$

$$a = 0,05 \text{ m}$$

$$k = 5\pi \text{ m}^{-1}$$

$$\omega = 20\pi \text{ s}^{-1}$$

$$\lambda = ? \checkmark$$

$$t_0 = ? \checkmark$$

$$c = ? \checkmark$$

$$\text{smer} = ?$$

$$y(0, 0) = ?$$

$$\ddot{y}(0, 0) = ?$$

$$\lambda = \frac{2\pi}{k} = \frac{2}{5} \text{ m}$$

$$c = \frac{\omega}{k} = \frac{20\pi \text{ m} \cdot \text{s}^{-1}}{5\pi \text{ s}^{-1}} = \underline{\underline{4 \text{ m/s}}}$$

$$V = \frac{c}{\lambda} = \frac{4 \text{ m/s}}{0,4 \text{ m}} = 10 \text{ Hz} \rightarrow t_0 = \frac{1}{10} \text{ s} = \underline{\underline{0,1 \text{ s}}}$$

$$\dot{y} = -\omega a \cos(-\frac{\pi}{4}) = -\omega a \frac{\sqrt{2}}{2}$$

$$\ddot{y} = -\omega^2 a \sin(-\frac{\pi}{4}) = \omega^2 a \frac{\sqrt{2}}{2}$$



Smer:

$$kx - \omega t \rightarrow \text{v smeri } x$$

$$kx + \omega t \leftarrow \text{v smeri } -x$$

$$-kx - \omega t \leftarrow \text{v smeri } -x$$

$$-kx + \omega t \rightarrow \text{v smeri } -(x)$$

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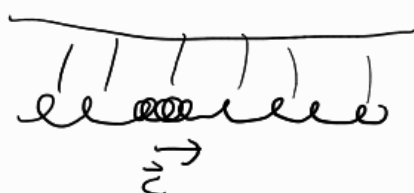
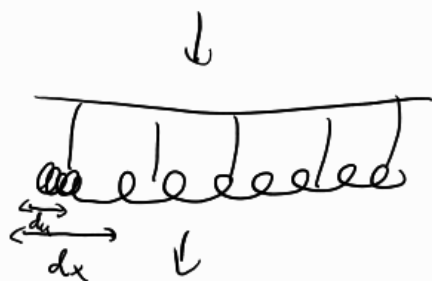
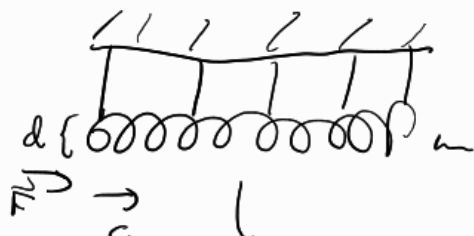
$$d = 10 \text{ cm}$$

$$m = 1 \text{ kg}$$

$$l = 6 \text{ m}$$

$$k = 100 \frac{\text{N}}{\text{m}}$$

$$c = ?$$



$$\begin{aligned} du &= v dt \\ dx &= c dt \end{aligned}$$

$$F dt = dm \cdot v = v \cdot m \cdot \frac{c dt}{l}$$

$$F = m v \frac{c}{l}$$

$$\begin{aligned} F &= k' du = \\ &= k \frac{l}{c dt} v dt = \\ &= k l \frac{v}{c} \end{aligned}$$

$$\begin{aligned} kl &= k' c dt \\ k' &= k \frac{l}{c dt} \end{aligned}$$

$$kl \frac{v}{c} = m v \frac{c}{l}$$

$$c^2 = \frac{kl^2}{m}$$

$$c = \sqrt{\frac{100 \cdot 36 \text{ m}^2 \text{ kg} \cdot \text{m}}{1 \text{ kg} \cdot \text{s}^2}} = \sqrt{3600} \frac{\text{m}}{\text{s}} = \underline{\underline{60 \frac{\text{m}}{\text{s}}}}$$

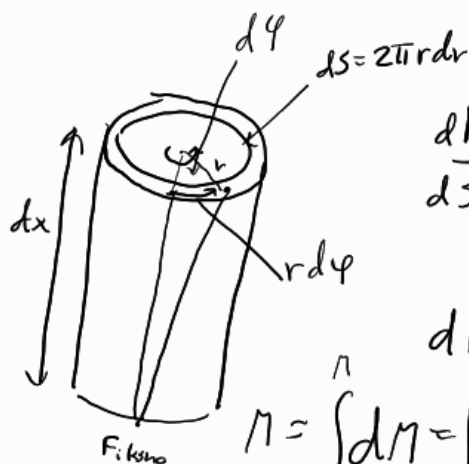
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$$l = 10 \text{ m}$$

$$\rho = 7.8 \frac{\text{g}}{\text{cm}^3}$$

$$G = 8 \cdot 10^9 \frac{\text{N}}{\text{mm}^2}$$

$$\varphi = ?$$



$$\frac{dF}{ds} = G \frac{r d\varphi}{dx}$$

$$dF = G \cdot 2\pi r^2 dr \frac{d\varphi}{dx}$$

$$M = \int_0^R dM = \int_0^R dF = G \cdot 2\pi \frac{d\varphi}{dx} \int_0^R r^3 dr = \underline{\underline{2\pi G \frac{d\varphi}{dx} \frac{R^4}{4}}}$$

Navar pri strigu spleznega valja

$$M = \underbrace{\frac{\pi}{2} G R^4 \frac{d\varphi}{dx}} = J\alpha \rightarrow dM = dJ\alpha = \frac{1}{2} d\pi R^2 \ddot{\varphi} = \frac{1}{2} R^2 \cdot \rho \cdot \pi R^2 dx \ddot{\varphi} =$$

$$= \frac{\pi}{2} \rho R^4 \ddot{\varphi} dx$$

$$\rightarrow \frac{dM}{dx} = \frac{\pi}{2} \rho R^4 \ddot{\varphi}$$

$$\cancel{\frac{\pi}{2} G R^4} \cdot \frac{d^2 \varphi}{dx^2} = \cancel{\frac{\pi}{2} \rho R^4} \frac{d^2 \varphi}{dt^2}$$

$$\frac{d^2 \varphi}{dx^2} = \frac{\rho}{G} \frac{d^2 \varphi}{dt^2}$$

$$\frac{1}{c^2} \rightarrow c = \sqrt{\frac{G}{\rho}}$$

$$t = \frac{l}{c} = \left(\sqrt{\frac{\rho}{G}} \right) = 10 \text{ m} \cdot \sqrt{\frac{7880 \frac{\text{kg}}{\text{m}^3} \cdot 3 \cdot 10^3 \frac{\text{Nm}}{\text{kg m}^2}}{8 \cdot 10^4 \frac{\text{Nm}}{\text{kg m}^2}}} = 3,1 \text{ ms}$$

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D-torzijski koeficient, $M = D \Delta \varphi$

$$D_a = 3 \cdot 10^3 \text{ Nm}$$

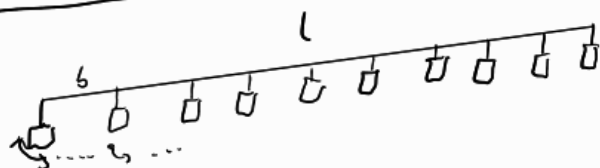
$$a = 1 \text{ m}$$

$$b = 10 \text{ m}$$

$$J = 30 \text{ kg m}^2$$

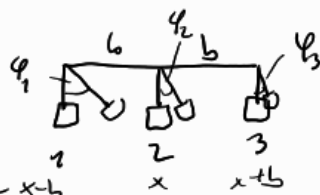
$$l = 100 \text{ m}$$

$$t = ?$$



$$D_a \propto \frac{1}{a}$$

$$D_b = \frac{D_a \cdot a}{b} = 3 \cdot 10^2 \text{ Nm}$$



$$M_{12} = D_b (\varphi_1 - \varphi_2)$$

$$M_{23} = D_b (\varphi_3 - \varphi_2)$$

$$M_{12} + M_{23} = J \ddot{\alpha}_2$$

$$D_b (\varphi_1 - 2\varphi_2 + \varphi_3) = J \ddot{\varphi}_2$$

$$\varphi_1 = \varphi(x-b)$$

$$\varphi_2 = \varphi(x)$$

$$\varphi_3 = \varphi(x+b)$$

$$\varphi(x+b) - 2\varphi(x) + \varphi(x-b) = \frac{J}{D_b} \ddot{\varphi}_2$$

$$\frac{\Delta \varphi}{\Delta x} = \frac{\varphi_{n+1} - \varphi_n}{\Delta x} = \varphi'_{n+\frac{1}{2}}$$

$$\frac{\Delta^2 \varphi}{\Delta x^2} = \frac{\Delta \left(\frac{\Delta \varphi}{\Delta x} \right)}{\Delta x} = \frac{\varphi'_{n+\frac{1}{2}} - \varphi'_{n-\frac{1}{2}}}{\Delta x} = \frac{\varphi_{n+1} - 2\varphi_n + \varphi_{n-1}}{\Delta x^2}$$

$$\frac{d^2 \varphi}{dx^2} = \left(\frac{J}{D_b b^2} \right) \frac{d^2 \varphi}{dt^2}$$

$$t = \frac{l}{c} = 3,16 \text{ s}$$

$$c = \sqrt{\frac{D_b b^3}{J}} = \sqrt{\frac{3 \cdot 10^2 \text{ Nm} \cdot 10^2 \text{ m}^3}{30 \text{ kg m}^2}} = 10^3 \frac{\text{m}}{\text{s}}$$

9.6

$$L = 10 \text{ cm}$$

$$T_1 = 20^\circ \text{C}$$

$$T_2 = 120^\circ \text{C}$$

$$t = ?$$

$$C^2 = \frac{\kappa R T}{M}$$

$$T(x) = (T_2 - T_1) \frac{x}{L} + T_1$$

$$C = \sqrt{\frac{\kappa R}{M} \sqrt{(T_2 - T_1) \frac{x}{L} + T_1}} = \frac{dx}{dt}$$

$$\int_0^t \sqrt{\frac{\kappa R}{M}} dt = \int_0^L \frac{1}{\sqrt{(T_2 - T_1) \frac{x}{L} + T_1}} dx$$

$$(T_2 - T_1) \frac{x}{L} + T_1 = u$$

$$(T_2 - T_1) \frac{1}{L} dx = du$$

$$t \cdot \sqrt{\frac{\kappa R}{M}} = \frac{L}{T_2 - T_1} \int_{T_1}^{T_2} u^{-\frac{1}{2}} du = \frac{L}{T_2 - T_1} 2\sqrt{u} \Big|_{T_1}^{T_2} = \frac{2L}{T_2 - T_1} (\sqrt{T_2} - \sqrt{T_1})$$

$$t = \frac{2L}{T_2 - T_1} \sqrt{\frac{M}{\kappa R}} (\sqrt{T_2} - \sqrt{T_1}) = 2,7 \cdot 10^{-3} \text{ s}$$

$$M = 29 \frac{\text{kg}}{\text{kmol}}$$

$$\kappa = 1,4$$

$$p = 8300 \frac{\text{kmol K}}{\text{kg}}$$

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$$l = 1 \text{ m}$$

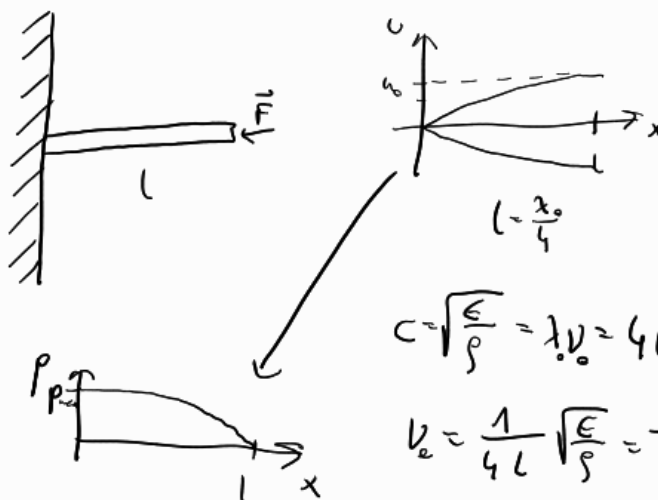
$$\rho = 8,9 \text{ g/cm}^3$$

$$E = 8 \cdot 10^5 \frac{\text{N}}{\text{cm}^2}$$

$$v_0 = ?$$

$$p(x=2 \text{ m}) = ?$$

$$s_0 = 3 \text{ mm}$$



$$C = \sqrt{\frac{E}{\rho}} = 1,4 v_0 = 46 v_0$$

$$v_0 = \frac{1}{46} \sqrt{\frac{E}{\rho}} = 750 \text{ Hz}$$

$$p = \frac{F}{s} = E \frac{\Delta l}{l}$$

Doppler

$$v_2 = v_1 \frac{c \pm v_2}{c \mp v_1}$$

1 - oddajnik

2 - sprejemnik

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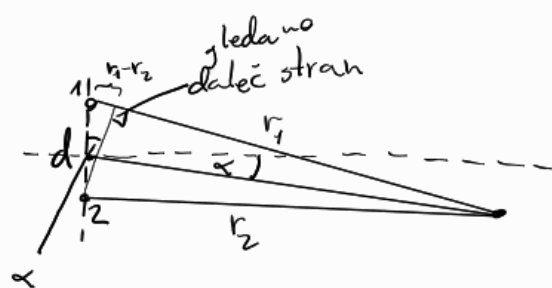
$$d = 12 \text{ cm}$$

$$\nu = 3 \text{ Hz}$$

$$\delta = \frac{2\pi}{4} = \frac{\pi}{2}$$

$$c = 25 \text{ cm/s}$$

$$\alpha_{\max}, \alpha_{\min}$$



$$s_1 = s_0 \sin(kr_1 - \omega t + \delta)$$

$$s_2 = s_0 \sin(kr_2 - \omega t)$$

$$s = s_1 + s_2 = s_0 (\sin(kr_1 - \omega t + \delta) + \sin(kr_2 - \omega t)) =$$

$$= 2s_0 \sin \frac{k(r_1 + r_2) - 2\omega t + \delta}{2} \cos \frac{k(r_1 - r_2) + \delta}{2} =$$

$$= 2s_0 \sin \left(k \frac{r_1 + r_2}{2} - \omega t + \frac{\delta}{2} \right) \underbrace{\cos \left(k \frac{r_1 - r_2}{2} + \frac{\delta}{2} \right)}_{\text{hi časovno odvisen}} \leftarrow \text{interferenčni člen}$$

$$r_1 - r_2 = d \sin \alpha$$

Pogoji za mini:

$$\cos \left(k \frac{r_1 - r_2}{2} + \frac{\delta}{2} \right) = 0$$

$$k \frac{r_1 - r_2}{2} + \frac{\delta}{2} = \frac{\pi}{2} + 2n\pi$$

$$k(r_1 - r_2) = \delta + 2n\pi$$

$$r_1 - r_2 = \frac{1}{k} (\delta + 2n\pi)$$

$$\sin \alpha_{\min} = \frac{c}{d 2\pi \nu} (\delta + 2n\pi) =$$

$$= \frac{c}{d \nu} \left(\frac{\delta}{2\pi} + n \right)$$

$$= \frac{c}{d \nu} \left(n + \frac{1}{4} \right)$$

$$\alpha_{\min} = \sin^{-1} \left(\frac{c}{d \nu} \left(n + \frac{1}{4} \right) \right)$$

$$k = \frac{\omega}{c}$$

pogoji za max: $\cos = \pm 1$

$$k \frac{r_1 - r_2}{2} + \frac{\delta}{2} = n\pi$$

$$r_1 - r_2 = (2n\pi - \delta) \frac{c}{2\pi \nu}$$

$$\sin \alpha_{\max} = \frac{c}{d 2\pi \nu} (2n\pi - \delta)$$

$$= \frac{c}{d \nu} \left(n - \frac{1}{4} \right)$$

$$\alpha_{\max} = \sin^{-1} \left(\frac{c}{d \nu} \left(n - \frac{1}{4} \right) \right)$$

n	0
$\alpha_{\min}$	
$\alpha_{\max}$	
...	